

LoggerAppsMsg()

Synopsis

```
int  LoggerAppsMsg(devProd, message_group, severity, message_string, ... )
int  devProd;
int  message_group;
int  severity;
char *message_string;
```

Description

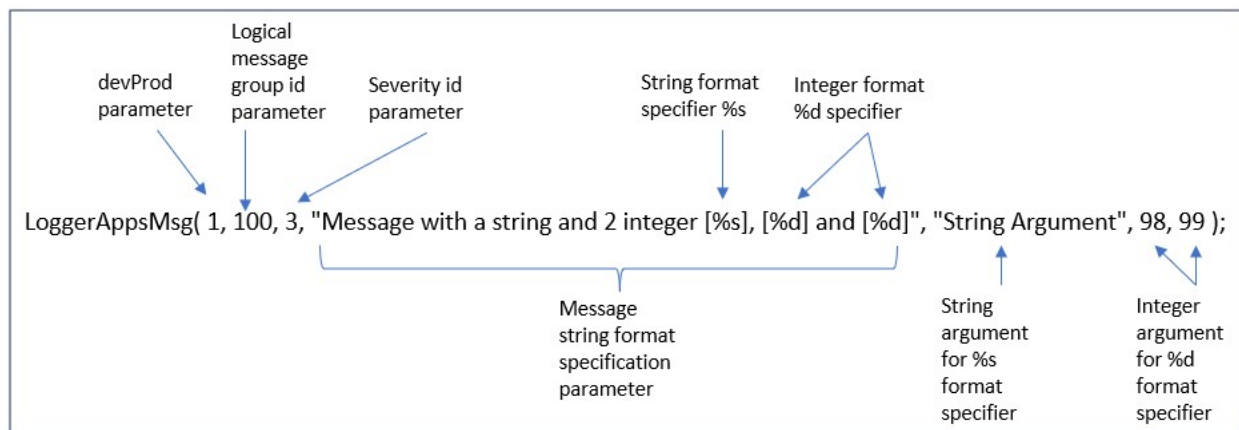
LoggerAppsMsg() decomposes argument parameters into id=value pairs, validates for errors, encode to reduce storage capacity, and securely protects to ensure confidentiality, integrity, and authenticity.¹

Example Usage

```
LoggerAppsMsg(1,100,3,"Message that generates 1 string and 2 integers [%s],
[%d] and [%d]","String Argument", 98,99);
```

Generates formatted output log message:

```
02/09/2026 12:01:55 [3650:sample_program] [DATABASE] [INFORM] Message that
generates 1 string and 2 integers [String Argument], [98] and [99]
```



The `devProd` integer parameter 1 indicates a development log message and 2 indicates a production log message. This will enable generating log messages 1 and 2 together or 2 only. The benefit of this is to prevent generating unnecessary development log messages in a production environment.

The logical `message_group` integer parameter is the id part of the id=value pairs defined in file `LoggerMessageDefines.txt`. The id part of 100 is translated and mapped to the corresponding value part of `DATABASE` in the output message. Refer to the `LoggerMessageDefines.txt` document for further details.

The `severity` integer parameter is the id part of the id=value pairs defined in file `LoggerMessageDefines.txt`. The id part of 3 is translated and mapped to the corresponding value part of

¹ The Free Version only supports decompose and validate format string specification and string argument parameters.

INFORM in the output message. Refer to the `LoggerMessageDefines.txt` document for further details.

The `message_string` format string specification parameter is a C-Programming Language `printf()` style format string specification that comprises of format string text and format specifiers. This `message_string` parameter is bounded by the guidelines in the `Wandi-SSAL-Bound-Capabilities.pdf` document.

The reminder of the `LoggerAppsMsg()` parameters (as indicated by "...") represents the corresponding format string specifiers argument values for `message_string`.

NOTE: Refer to the `Wandi-SSAL-Bound-Capabilities.pdf` document for validation guidelines used by `LoggerAppsMsg()` to check and validate parameter values.

`LoggerAppsMsg()` returns 0 on success and -1 on error.

This API function is failed safe which means when an error occurs the application can continue processing.

Error Codes

Error codes are written to the following files, where the `<cpid>` is the application process id number:

- `/tmp/<cpid>-LoggerAppsMessage.txt`
- `$LOGGER_HOME_PATH/LoggerOutput/<cpid>-LoggerAppsMessage.txt`

135	Invalid <code>devProd</code> parameter. Solution: Refer to <code>devProd</code> guidelines above.
136	Invalid <code>message_group</code> , <code>severity</code> and <code>message_string</code> parameter. Solution: Refer to <code>message_group</code> and <code>message_string</code> guidelines above.
137	Invalid <code>message_string</code> parameter. Solution: Refer to <code>message_string</code> guidelines above
138	Too many format string type specifiers. Solution: Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.
139	String argument pointer parameter is NULL. Solution: Make sure string argument pointer is not NULL.
140	String argument pointer is -1. Solution: Make sure string argument pointer is not -1.
141	String argument not properly NULL terminated. Solution: Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.
142	String argument exceeded allowed length of 64 characters. Solution: Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.
143	Invalid string argument value. Solution: Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.
144	Invalid format string specifier type. Solution: Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.
115, 117	Parameter <code>message_string</code> is NULL or exceeded allowed length of 256 characters. Solution: Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.
104, 106, 116	Invalid ASCII character in <code>message_string</code> or string argument.

	Solution: Refer to <code>Wandi-SSAL-Bound-Capabilities.pdf</code> guidelines.
105	String argument length is zero. Solution: Make sure string argument is not zero length.